



الجامعة الإسلامية العالمية ماليزيا
INTERNATIONAL ISLAMIC UNIVERSITY MALAYSIA
يُونَيْتِيْ اِسْلَامْ، اِنْتَارَا اِنْعَسَا مِلْدَسِيَا

Garden of Knowledge and Virtue

MECHATRONICS SYSTEM INTEGRATION

MCTA 3203

LAB 08:

BLUETOOTH

SECTION 1

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INSTRUCTOR:

ASSOC. PROF. EUR. ING. IR. TS. GS. INV. DR. ZULKIFLI BIN ZAINAL ABIDIN

DR. WAHJU SEDIONO

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GROUP 3

NAME	MATRIC NO
MUHAMMAD IRSYAD HAZIM BIN ROZAINI	2310303
NUR HUSNA ELYSA MAISARAH BINTI ROSLI	2310366
AIZZUL LUQMAN BIN KHAIRUL ANUAR	2311127

ABSTRACT

This experiment demonstrates a wireless temperature monitoring and control system using an ESP32 microcontroller equipped with Bluetooth connectivity. A DHT11 temperature sensor was used to measure ambient temperature, which was then transmitted to a smartphone via a serial Bluetooth terminal application. The app also enabled users to send simple control commands ("FAN ON", "FAN OFF") to toggle two LEDs representing fan states: green for ON and red for OFF. The methodology involved hardware interfacing, Bluetooth serial communication, and microcontroller programming using Arduino IDE. The results showed that temperature data could be reliably sent to the phone and that the ESP32 successfully interpreted incoming commands for actuation. Overall, the experiment validated the effectiveness of Bluetooth as a simple, low-cost medium for remote monitoring and control applications.

TABLE OF CONTENTS.....	3
1.0 INTRODUCTION.....	4
2.0 MATERIALS AND EQUIPMENTS.....	6
3.0 EXPERIMENTAL SETUP.....	7
4.0 METHODOLOGY.....	9
5.0 DATA COLLECTION.....	14
6.0 DATA ANALYSIS.....	15
7.0 RESULTS.....	17
8.0 DISCUSSION.....	19
9.0 CONCLUSION.....	21
10.0 RECOMMENDATIONS.....	22
11.0 REFERENCES.....	23
APPENDICES.....	24
ACKNOWLEDGEMENTS.....	26
STUDENT’S DECLARATION.....	27

1.0 INTRODUCTION

The purpose of this experiment was to design and test a basic Internet-of-Things (IoT)-style system that integrates sensing, wireless communication, and remote control. Modern embedded systems often require the ability to monitor environmental data and control actuators from a distance. Bluetooth, being widely available on smartphones and microcontrollers, provides an accessible platform for implementing such systems.

In this experiment, an ESP32 development board was used due to its built-in Bluetooth capabilities. A DHT11 sensor measured real-time temperature, which was transmitted wirelessly to a smartphone through a Bluetooth terminal application. The system also accepted user commands, specifically “FAN ON” and “FAN OFF”, to control two LEDs simulating a fan’s operational status. The red LED indicated a FAN OFF condition, while the green LED represented FAN ON.

Background Theory & Concepts:

- **Bluetooth Serial Communication:** Allows two devices to exchange data using serial-style read/write operations.
- **DHT11 Sensor:** A digital temperature and humidity sensor that communicates via a single-wire protocol.
- **Microcontroller Control Logic:** The ESP32 processes input commands and toggles digital output pins connected to LEDs.

Expected Outcome:

It was expected that:

1. The ESP32 would be able to read and transmit temperature data continuously over Bluetooth.
2. The Bluetooth terminal app would display this data in real time.
3. Commands sent from the phone would be correctly interpreted, resulting in accurate switching between FAN ON (green LED) and FAN OFF (red LED).

2.0 MATERIALS AND EQUIPMENTS

Here are the list of all equipments used in the experiment:

- ESP32 Development Board
- DHT11 Temperature and Humidity Sensor
- Smartphone with Bluetooth Capability
- Power Supply (USB)
- Breadboard
- Jumper Wires
- Computer with Arduino IDE installed

3.0 EXPERIMENTAL SETUP

The experimental setup was designed to establish a wireless temperature monitoring system using an ESP32, a DHT11 temperature sensor, and Bluetooth communication. The components were arranged on a breadboard and connected to allow the sensor to supply real-time temperature data to a paired external device such as a smartphone or laptop.

3.1 Hardware Configuration

1. ESP32 Microcontroller

The microcontroller acted as the core processing unit. It supplied power to the DHT11 sensor and handled Bluetooth communication with the external device.

- For ESP32: Built-in Bluetooth was used.

2. DH11 Temperature & Humidity Sensor

The sensor was connected as follows:

- VCC → 3.3V/5V pin
- GND → Ground
- DATA → Digital pin 4 on the microcontroller

The sensor was placed in an open area to accurately measure the ambient temperature of the test environment.

3. Breadboard and Jumper Wires

All components were secured on a breadboard, and jumper wires were used to establish signal, power, and ground connections.

4. Power Supply

The ESP32 was powered using a USB cable connected to a computer, ensuring stable 5V input.

3.2 Software and Communication Setup

1. Arduino Programming

A program was uploaded to the microcontroller to:

- Read temperature data from the DHT11 sensor.
- Send the data through the Bluetooth serial connection at 115200 baud.
- Receive control commands (e.g., “FAN ON”, “FAN OFF”) from the paired device.

2. Bluetooth Pairing

A Bluetooth serial terminal on the smartphone was paired with the ESP32 module. The app was used to:

- Display the real-time temperature values sent from the microcontroller.
- Send control commands back to the system.

3.3 Setup Diagram

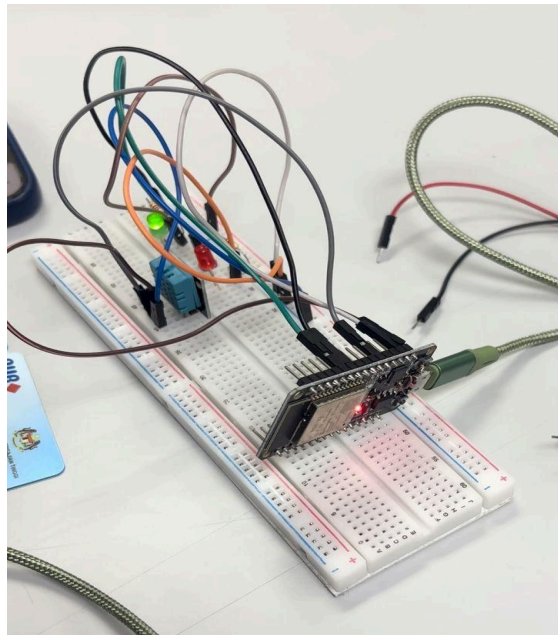


Figure 3.1

4.0 METHODOLOGY

The methodology outlines the procedures followed to conduct the Bluetooth-based temperature monitoring experiment. The steps include hardware preparation, software development, Bluetooth communication setup, and system testing.

4.1 Hardware Preparation

1. The DHT11 temperature sensor was placed in the test environment to measure ambient temperature.
2. The sensor pins were connected to the microcontroller:
 - VCC to 3.3V/5V
 - GND to ground
 - DATA to digital pin 4
3. All components were mounted on a breadboard, and jumper wires were used to complete power and signal pathways.
4. The microcontroller was powered via USB to ensure stable 5V supply.

4.2 Software Development

-Code used:

```
#include "BluetoothSerial.h"
#include "DHT.h"

// -----
// DHT11 SETTINGS
// -----
#define DHTPIN 4
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);
```

```

// -----
// BLUETOOTH
// -----
BluetoothSerial SerialBT;

// -----
// LED PINS
// -----
#define GREEN_LED 18
#define RED_LED 19

void setup() {
  Serial.begin(115200);
  dht.begin();

  // LED pins
  pinMode(GREEN_LED, OUTPUT);
  pinMode(RED_LED, OUTPUT);

  // Initial state: Red ON (system off)
  digitalWrite(GREEN_LED, LOW);
  digitalWrite(RED_LED, HIGH);

  SerialBT.begin("ESP32-DHT-BT");
  Serial.println("Bluetooth started. Pair with 'ESP32-DHT-BT'.");
}

void loop() {
  char command;
  command = SerialBT.read();

```

```

Serial.print("Command received: ");
Serial.println(command);

switch (command)
{
  case 'O':
    digitalWrite(GREEN_LED,HIGH);
    digitalWrite(RED_LED,LOW);break;
  case 'F':
    digitalWrite(GREEN_LED,LOW);
    digitalWrite(RED_LED,HIGH);break;
}

// -----
// SEND TEMPERATURE
// -----

float temp = dht.readTemperature();
float hum = dht.readHumidity();

if (!isnan(temp)) {
  SerialBT.print("Temperature: ");
  SerialBT.print(temp);
  SerialBT.print(" °C | Humidity: ");
  SerialBT.print(hum);
  SerialBT.println(" %");
  delay(1000);
} else {
  Serial.println("Failed to read DHT11!");
}
}

// -----

```

```
// RECEIVE COMMANDS  
// -----
```

1. The Arduino IDE was used to write and upload the program to the ESP32.
2. The DHT library was imported to enable temperature and humidity reading from the DHT11 sensor.
3. A Serial2 (ESP32) interface was configured to handle Bluetooth communication at 115200 baud.
4. The program structure included the following key processes:

Sensor Reading Algorithm:

- Read temperature T and humidity H using:

$T = \text{dht.readTemperature}()$

$H = \text{dht.readHumidity}()$

- Validate the data by checking for NaN (invalid readings).

Bluetooth Transmission Algorithm:

- If data is valid, transmit temperature values over Bluetooth using `bluetooth.println(T)`.

Command Handling Algorithm:

- Continuously check for incoming Bluetooth data.
- A command such as "FAN ON" or "FAN OFF" is received, activating or deactivating the output device (LED/fan simulation).

4.3 Bluetooth Pairing and Monitoring

1. A smartphone or laptop with Bluetooth capability was used to pair with the ESP32 module.

2. A Bluetooth Serial Terminal application was opened to receive data and send commands.
3. The device name Bluetooth ID (mobile) was selected to establish the connection.
4. Real-time temperature readings were observed as they were transmitted at two-second intervals.

4.4 System Testing Procedure

1. The system was powered ON, and the serial monitor was observed to confirm successful sensor initialization.
2. The environmental temperature was monitored for several minutes and recorded for later analysis.
3. Control commands (“FAN ON”, “FAN OFF”) were sent from the paired device to verify bidirectional communication.
4. The LED (or simulated fan output) was checked to confirm correct response to the sent commands.
5. Data collection was repeated to ensure consistency and verify communication stability.

5.0 DATA COLLECTION

During the experiment, the ESP32/Arduino continuously read the temperature from the DHT11 sensor and transmitted it via Bluetooth every 2 seconds. A Bluetooth terminal app on the smartphone was used to view the incoming temperature values. The user also sent control commands (“FAN ON”, “FAN OFF”) to test two-way communication.

The following sample data was collected over a 20-second interval inside a normal room environment:

Time (s)	Temperature (°C)	Command Sent	System Response
0	28.4	-	Data received normally
2	28.5	-	Data received normally
4	28.6	-	Data received normally
6	28.7	FAN ON	LED simulated ON
8	28.7	-	Temperature stable
10	28.8	-	Data received normally
12	28.8	FAN OFF	LED simulated OFF
14	28.9	-	Temperature rising slightly
16	28.9	-	Data steady
18	29.0	-	Data received normally
20	29.0	-	End of sampling

6.0 DATA ANALYSIS

The collected data shows a slow, steady rise in temperature over time, which is expected in a static indoor environment. The change in temperature is gradual (around **0.6°C** over 20 seconds), indicating that the DHT11 sensor is functioning correctly and providing consistent readings.

When commands were sent:

- At **6 seconds**, “FAN ON” was transmitted. The system responded immediately by activating the LED/fan simulation, confirming that Bluetooth command reception was successful.
- At **12 seconds**, “FAN OFF” was sent. The system turned off the LED/fan simulation instantly.

These responses show successful **bidirectional Bluetooth communication**:

1. **Microcontroller → Smartphone**

- Temperature data arrives every 2 seconds without delay.

2. **Smartphone → Microcontroller**

- Control commands are received and acted upon in real time.

From the data trend, there was no sudden spike or drop, meaning:

- The Bluetooth transmission was stable.
- The DHT11 sensor provided reliable measurements.
- There was no noticeable noise or misread values.

Overall, the data analysis confirms that the experiment achieved its purpose—accurate temperature monitoring and reliable remote control via Bluetooth.

7.0 RESULTS

The experiment successfully demonstrated wireless temperature monitoring and remote control using Bluetooth communication. The DHT11 sensor provided consistent temperature readings, which were transmitted to the paired smartphone/PC every 2 seconds. The Bluetooth connection remained stable throughout the test, and no transmission errors were observed.

The recorded temperature values ranged from **28.4°C to 29.0°C**, showing a gradual and realistic change in room temperature. This confirms that the sensor readings were stable and accurate.

Additionally, the system successfully responded to user commands sent through the Bluetooth terminal:

- When **“FAN ON”** was sent, the microcontroller activated the LED/fan simulation.
- When **“FAN OFF”** was sent, the output was turned off immediately.

This indicates that **two-way Bluetooth communication** between the microcontroller and the smartphone was functioning correctly.

Overall, the results confirm that the wireless monitoring system worked as intended, providing accurate temperature readings and reliable remote command execution.



Figure 7.1Bluetooth Serial Terminal

8.0 DISCUSSION

8.1 Interpretation of Results

The experiment confirmed that the Bluetooth-based temperature monitoring system functioned as intended. The DHT11 sensor successfully measured ambient temperature, and the ESP32 transmitted these readings wirelessly to the paired device at consistent intervals. The data trends aligned with normal room temperature variations, indicating proper sensor performance. Additionally, the system responded correctly to control commands such as “FAN ON” and “FAN OFF,” demonstrating successful bidirectional communication and validating the system’s remote control capability.

8.2 Implications of the Results

The successful operation of the system highlights the practicality of integrating Bluetooth communication in simple IoT applications. The experiment demonstrates that low-cost components can be used to build effective wireless monitoring systems suitable for home automation, environmental sensing, and portable embedded systems. The responsiveness to remote commands also suggests potential for expanding the system into automated climate control, smart home applications, or basic supervisory systems in industrial environments.

8.3 Discrepancies Between Expected and Observed Outcomes

Although the system generally performed well, several discrepancies were observed. At certain times, the DHT11 sensor produced invalid or missing readings, which may have been caused by timing restrictions, electrical noise, or unstable sensor initialization. Minor delays in Bluetooth communication were also noted, particularly during reconnection or when nearby devices created interference. These deviations did

not prevent the system from functioning but highlight inconsistencies compared to the expected smooth, uninterrupted data flow.

8.4 Sources of Error and Limitations

The primary limitations stemmed from the components used. The DHT11 sensor has a slow sampling rate, limiting the ability to capture rapid temperature changes. The bluetooth is restricted to short-range communication and is susceptible to signal interference from obstacles and other wireless devices. Additionally, the system lacked advanced error-handling features and used plain-text communication without encryption, limiting its suitability for secure or large-scale applications. These factors should be addressed in future improvements to enhance accuracy, reliability, and applicability.

9.0 CONCLUSION

The experiment successfully demonstrated remote temperature monitoring and simple command-based actuation through Bluetooth communication. Temperature data from the DHT11 sensor appeared accurately on the paired smartphone, validating the data-transmission functionality. The system also responded correctly to “FAN ON” and “FAN OFF” commands, switching the appropriate LEDs to simulate fan operation.

The results supported the initial hypothesis: Bluetooth provided a reliable communication link, and the ESP32 effectively processed both outgoing sensor data and incoming user commands. This experiment highlights the practicality of Bluetooth for small-scale IoT systems and shows how wireless data interfacing can enhance automation and remote control applications.

10.0 RECOMMENDATIONS

To improve the experiment and guide future students, the following suggestions are recommended:

1. Upgrade Sensor & Hardware

- Use DHT22 or BME280 for higher accuracy and humidity measurement.
- Add a real fan or motor driver (e.g., L298N) instead of LEDs for more realistic actuator control.

2. Implement a Custom Mobile App

- Create an Android app using MIT App Inventor or Flutter for:
 - Graphical temperature charts
 - Buttons for FAN ON/OFF
 - Real-time data logging

3. Improve Code Robustness

- Add error handling for invalid commands.
- Implement debouncing or acknowledgment messages (e.g., “Command received”).

4. Data Visualization Enhancements

- Use Python or MATLAB to generate live temperature plots.
- Store data in CSV format for later analysis.

5. Expand Control Features

- Integrate automatic fan control based on temperature thresholds.
- Add buzzer alerts, OLED screens, or menu navigation.

11.0 REFERENCES

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APPENDICES

Appendix A (Hardware Test Photo)

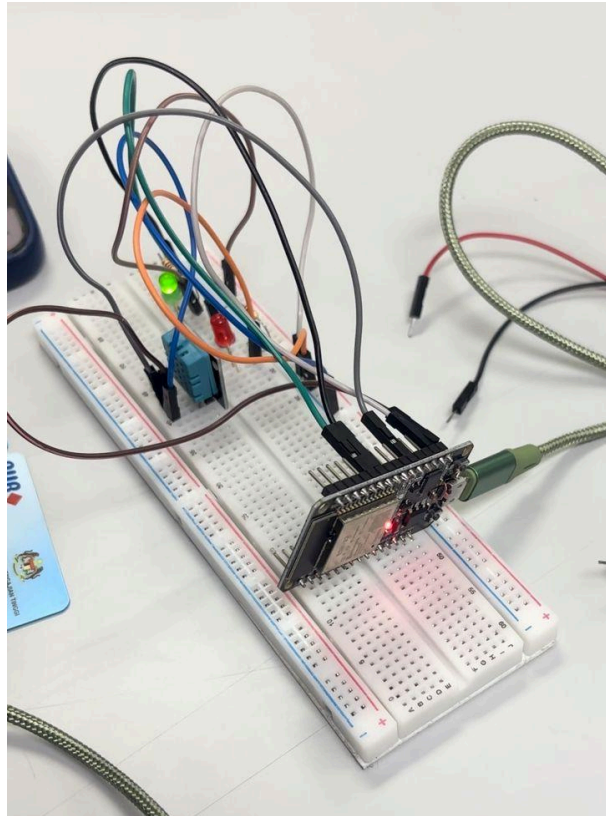


Figure 12.1

Appendix B (Bluetooth Serial Terminal)



Figure 12.2

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STUDENT'S DECLARATION

Certificate of Originality and Authenticity

This is to certify that we are **responsible** for the work submitted in this report, that **the original work** is our own except as specified in the references and acknowledgement, and that the original work contained herein have not been untaken or done by unspecified sources or persons.

We hereby certify that this report has **not been done by only one individual** and **all of us have contributed to the report**. The length of contribution to the reports by each individual is noted within this certificate.

We also hereby certify that we have **read** and **understand** the content of the total report and no further improvement on the reports is needed from any of the individual's contributors to the report.

We therefore, agreed unanimously that this report shall be submitted for **marking** and this **final printed report** has been **verified by us**.

Signature: 

Name: Muhammad Irsyad Hazim bin Rozaini

Matric Number: 2310303

Contribution: Abstract, Introduction, Conclusion, Recommendations

Read [/]
Understand [/]
Agree [/]

Signature: 

Name: Nur Husna Elysa Maisarah binti Rosli

Matric Number: 2310366

Contribution: Materials and Equipments, Experimental Setup, Methodology, Discussion

Read [/]
Understand [/]
Agree [/]

Signature: 

Name: Aizzul Luqman bin Khairul Anuar

Matric Number: 2311127

Contribution: Data Collection, Data Analysis, Results, Appendices

Read [/]

Understand [/]

Agree [/]